

R. Kyle Bocinsky, PhD, RPA

Contact Information

Montana Climate Office
University of Montana
Missoula, MT 59812

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Research Interests

*human dimensions of climate change • computational social science • paleo-environments
complex systems • plant and animal domestication • Indigenous ecologies • reproducibility*

Education

Washington State University, Pullman, Washington
PhD, Anthropology, December 2014
Dissertation title: *Landscape-based Null Models for Archaeological Inference*
Committee: Timothy Kohler (chair), Jade d'Alpoim Guedes, Andrew Duff, and Colin Grier

MA, Anthropology, May 2011
Thesis title: *Is a Bird in Hand Really Worth Two in the Bush? Models of Domestication on the Colorado Plateau*
Committee: Timothy Kohler (chair), Andrew Duff, Donna Glowacki, and Brian Kemp

University of Notre Dame, Notre Dame, Indiana
Glynn Family Honors Program, 2004–2008
BA, Anthropology, May 2008
Thesis title: *Rodent Stable Carbon-isotope Ratios as a Measure of Maize Production*
Advisor: Mark Schurr

Appointments

University of Montana, Missoula, Montana
Jul 2022–Present Assistant Research Professor,
Department of Society and Conservation,
W.A. Franke College of Forestry and Conservation
Jan 2021–Present Director of Climate Extension, Montana Climate Office
Jul 2024–Dec 2024 Faculty Fellow, Experiential Learning and Career Success
Aug 2020–Dec 2020 Research Affiliate, Montana Climate Office
Jan 2018–Jul 2020 Research Associate, Montana Climate Office

Desert Research Institute, Reno, Nevada
June 2017–Present Hourly Assistant Research Professor,
Division of Earth and Ecosystem Sciences

Crow Canyon Archaeological Center, Cortez, Colorado
Jan 2021–Feb 2025 Research Associate
Aug 2018–Dec 2020 Director of the Research Institute
William D. Lipe Chair in Research
Jan 2017–July 2018 Research Associate
Jan 2016–Dec 2016 Director of Sponsored Projects

Washington State University, Pullman, Washington
Jan 2016–Dec 2020 Adjunct Research Faculty, Department of Anthropology
Jan 2015–Dec 2015 Post-doctoral Researcher, Department of Anthropology
Aug 2014–Dec 2014 Teaching Assistant, Department of Anthropology
Aug 2009–Jul 2014 Research Fellow, Department of Anthropology
Aug 2008–Jul 2009 Research Assistant, Department of Anthropology

Grants

- 2025–
2027 **Co-PI, NSF R212: The Northern Great Plains Regional Incubator for Drought Resiliency**
Program: National Science Foundation, Division of Research, Innovation, Synergies and Education (GEO/RISE), Integrative and Collaborative Education and Research (ICER), *Regional Resilience Innovation Incubator (R212)*
Award: 2519681
Amount: \$499,927 awarded to UM
PI: Kelsey Jencso, University of Montana
- 2025–
2027 **Co-I, Očhéthi Šakówiŋ Data for Climate Resilience**
Program: Bureau of Indian Affairs, Branch of Tribal Community Resilience
Award:
Amount: \$248,818 total
PI: Janna Black, Great Plains Tribal Water Alliance
- 2022–
2027 **Co-PD (UM PD), Native Climate: Strengthening the role of Climate Hubs in Indian Country**
Program: United States Department of Agriculture, National Institute of Food and Agriculture, *Extension, Education & USDA Climate Hubs Partnership*
Award: 2022-68015-36357
Amount: \$1,500,000 total, \$291,981 awarded to UM
PD: Maureen McCarthy, Desert Research Institute
- 2023–
2026 **PI, Enhancing Climate-smart Disaster Relief in FSA Programs: Non-stationarity at the Intersection of Normal Grazing Periods and US Drought Assessment**
Program: United States Department of Agriculture, Agricultural Research Service, *Research, Education, and Economics*
Award: 58-3070-3-016
Amount: \$225,000 awarded to UM
- 2023–
2025 **Co-PI (UM PI), Collaborative Research: HNDS-R: Human Networks, Sustainable Development, and Lived Experience in a Nonindustrial Society**
Program: National Science Foundation, Division of Behavioral and Cognitive Sciences, Directorate for Social, Behavioral & Economic Sciences, *Human Networks & Data Science – Research*
Award: 2212725
Amount: \$498,706 total, \$25,393 awarded to UM
PI: Scott Ortman, University of Colorado, Boulder
- 2023–
2025 **PI, Native Drought Resilience: CSKT Actions for Climate-Drought Adaptation**
Program: National Oceanic & Atmospheric Administration, National Integrated Drought Information System, *Coping with Drought – Building Tribal Drought Resilience*
Award: NA22OAR4310232
Amount: \$498,194 awarded to UM, including sub-awards to the Confederated Salish and Kootenai Tribes of the Flathead Reservation and Salish Kootenai College
- 2023–
2025 **Co-I, Upper Missouri River Basin Project: Plains Snow and Soil Moisture Monitoring Network – Operations and Maintenance**
Program: National Oceanic & Atmospheric Administration, National Mesonet Program
Award: 1305M223PNWWG0302
Amount: \$1,696,229 awarded to UM
PI: Kelsey Jencso, University of Montana
- 2022–
2023 **PD, Disaster Resilience Planning Assistance and Hazard Mitigation Planning Fort Peck Assiniboine and Sioux Tribes**
Program: Environmental Protection Agency, *Pre Disaster Mitigation Grant Program; American Rescue Plan*
Award:
Amount: \$40,000 awarded to UM through MOU with Spackman Massop Michaels

- 2022–
2023 **Co-I, Developing the NIDIS Drought Early Warning System for the Upper Missouri River Basin**
Program: National Oceanic & Atmospheric Administration, National Integrated Drought Information System, *Climate and Atmospheric Research*
Award: NA21OAR4310473
Amount: \$344,568 awarded to UM
PI: Kelsey Jencso, University of Montana
- 2021–
2023 **Co-PD (UM PD), TCU Missouri River Watershed Consortium**
Program: United States Department of Agriculture, National Institute of Food and Agriculture, *Tribal Colleges Research Grants Program*
Award: 2021-38424-34953
Amount: \$220,000 total, \$64,508 awarded to UM
PD: Ruth Plenty Sweetgrass-She Kills, Nueta Hidatsa Sahnish College
- 2021–
2023 **Co-PD (UM PD), All Climate is Local: 'Multiple ways of knowing' to enhance resilience of tribal agroecosystems to droughts, floods, and rising temperatures**
Program: United States Department of Agriculture, National Institute of Food and Agriculture
Award: 2021-67019-33420
Amount: \$49,998 awarded to DRI
PD: Maureen McCarthy, Desert Research Institute
- 2020–
2023 **Co-PD, COVID-19 Rapid Response Toolkit for Tribal Extension Agents**
Program: United States Department of Agriculture, National Institute of Food and Agriculture
Award: 2020-68008-32762
Amount: \$300,000 awarded to DRI/CCAC
PD: Maureen McCarthy, Desert Research Institute
- 2020–
2022 **Co-PD, Teaching Native Waters: Place-based Community of Practice for STEM Teachers Serving Native American Students**
Program: United States Department of Agriculture, National Institute of Food and Agriculture, *Women and Minorities in Science, Technology, Engineering, and Mathematics* (WAMS) program
Award: 2020-38503-31952
Amount: \$99,998 awarded to DRI
PD: Meghan Collins, Desert Research Institute
- 2019–
2022 **Project Manager, The Maize Database Project**
Program: History Colorado – State Historical Fund
Award: 2019-M2-001
Amount: \$34,912 awarded to CCAC
PI: Mark Varien, Crow Canyon Archaeological Center
- 2018–
2022 **Co-PD, Improving the Efficacy of Climate Information for Water Use Decisions: Developing, Testing, and Institutionalizing New Tools for Producers**
Program: United States Department of Agriculture, National Institute of Food and Agriculture, Agriculture and Food Research Initiative program area special competition, *Water for Agriculture: Understanding Decisions and Behaviors Connected with Agriculture and Post-harvest Processing Industry Water Use*
Award: 2017-67027-26313
Amount: \$462,539 awarded to UM
PD: Laurie Yung, University of Montana

- 2016–
2022 **Co-PI, Developing and Deploying SKOPE—A resource for Synthesizing Knowledge of Past Environments**
Program: National Science Foundation special competition, *Resource Implementations for Data Intensive Research in the Social Behavioral and Economic Sciences*
Award: 1637171
Amount: \$1,339,658 total, \$254,189 awarded to WSU/CCAC
PI: Tim Kohler, Washington State University
- 2015–
2022 **Co-PD, Native Waters on Arid Lands: Enhancing Climate Resiliency & Agriculture on American Indian Land**
Program: United States Department of Agriculture, National Institute of Food and Agriculture, Agriculture and Food Research Initiative program area special competition, *Water for Agriculture*
Award: 2015-69007-23190
Amount: \$2,516,844 awarded to the University of Nevada, Reno
PD: Maureen McCarthy, University of Nevada, Reno
- 2010–
2015 **Fellow, NSF Graduate Research Fellowship Program**
Program: National Science Foundation
Award: 1347973
Amount: \$92,000 stipend and \$32,000 cost-of-education allowance over three years
PI: Tim Kohler, Washington State University
- 2009–
2011 **Trainee, IGERT Program in Evolutionary Modeling**
Program: National Science Foundation
Award: 0549425
Amount: \$60,000 stipend and \$20,000 cost-of-education allowance over two years
PI: Tim Kohler, Washington State University

Awards & Scholarships

- 2022 **Presidential Recognition Award**, Society for American Archaeology
Awarded to members of the SAA Bylaws Committee (on which Bocinsky served) for their thorough and thoughtful work updating the SAA Bylaws, the foundation of the Society's governing documents
- 2015 **Cordell Prize**, Pecos Conference, Mancos, Colorado
Awarded for the best extemporaneous talk presented by an archaeologist 35 years of age or younger
- 2014 **Graduate Achievement Award in the Social Sciences at the Doctoral Level**, College of Arts and Sciences, Washington State University
Awarded to a doctoral student for outstanding academic performance, achievements in research, scholarly, and/or creative activity, and contributions or service to the social sciences, the college and the institution during their advanced degree studies at Washington State University
- 2013 **R.E. Taylor Student Poster Award**, Society for Archaeological Science
Awarded for the best student research poster presented at the Society for American Archaeology meetings displaying innovative contributions to archaeological research through the use of scientific methods
- 2011 **Elaine Burgess Graduate Fellowship in Anthropology**, Department of Anthropology, Washington State University
Awarded to a graduate student of outstanding professional promise in anthropology without consideration of subfield

- 2009 **Nicholas Michael Scoales Scholarship in Archaeology**, Department of Anthropology, Washington State University
Awarded to provide research support for an outstanding graduate student in archaeology specializing in analysis of lithic artifacts or in western North American prehistory
- 2008 **The Rev. Raymond W. Murray, C.S.C. Award in Anthropology**, Department of Anthropology, University of Notre Dame
Awarded to the top graduating senior for exemplary work in anthropology
- Peer-reviewed Articles**
-  open access
- 2025 Yegorova, Svetlana V., Solomon Z. Dobrowski, Laurie Yung, Sean A. Parks, **R. Kyle Bocinsky**, Kimberley T. Davis, Caitlin Littlefield, Marco P. Maneta, Carina Wyborn, Patrick Wurster, Robin Rank, Douglas Brinkerhoff, and Thomas Colligan
Spatial climate analogs in climate change research, impact assessment, and decision-making. *BioScience* 75: 362–378.
DOI: <https://doi.org/10.1093/biosci/biaf031>
-  2025 Kohler, Timothy A., Laura E. Brumbaugh, and **R. Kyle Bocinsky**
How many people lived in the Chaco Regional System, and why it matters. *KIVA*: 1–31.
DOI: <https://doi.org/10.1080/00231940.2025.2553441>
Code: <https://github.com/bocinsky/kohler2025>
-  2025 Lisonbee, Joel, Britt Parker, Erica Fleishman, Trent W Ford, **R. Kyle Bocinsky**, Gretel Follingstad, Abby G Frazier, Zachary H Hoylman, Amy R. Hudson, John W. Nielsen-Gammon, Natalie A. Umphlett, Elliot Wickham, Aparna Bamzai-Dodson, Royce Fontenot, Brian Fuchs, John Hammond, Jeffrey E. Herrick, Mike Hobbins, Andrew Hoell, Jacob Jones, Erin Lane, Zachary Leasor, Yongqiang Liu, Jason A. Otkin, Amanda Sheffield, Dennis Todey, and Roger Pulwarty
Prioritization of Research on Drought Assessment in a Changing Climate. *Earth's Future* 13(3): e2024EF005276.
DOI: <https://doi.org/10.1029/2024EF005276>
- 2024 Snitker, Adam J., Laurie Yung, Elizabeth Covelli Metcalf, **R. Kyle Bocinsky**, Neva Hassanein, Kelsey Jencso, Ada P. Smith, Austin Schuver
How agricultural producers use local knowledge, climate information, and on-farm “experiments” to address drought risk. *Agriculture and Human Values* 41: 1857–1875.
DOI: <https://doi.org/10.1007/s10460-024-10582-3>
-  2024 Hoylman, Zachary H., Zach Holden, **R. Kyle Bocinsky**, David Ketchum, Alan Swanson, and Kelsey Jencso
Optimizing Drought Assessment for Soil Moisture Deficits. *Water Resources Research* 60(6): e2023WR036087.
DOI: <https://doi.org/10.1029/2023WR036087>
Code: <https://github.com/zhoylman/soil-moisture-validation>
- 2024 Bird, Darcy, **R. Kyle Bocinsky**, Kelsey M. Reese and Timothy A. Kohler
Settlement persistence in the prehispanic central Mesa Verde Region: A dynamic analysis. *Journal of Archaeological Science: Reports* 56: 104515.
DOI: <https://doi.org/10.1016/j.jasrep.2024.104515>
Code: <https://github.com/darcybird/settlement-persistence-in-CMV>

- 2024 Gillreath-Brown, Andrew D., **R. Kyle Bocinsky**, and Timothy A. Kohler
A low-frequency summer temperature reconstruction for the United States Southwest, 3000 BC–AD 2000. *The Holocene* 34(4): 451–466.
DOI: <https://doi.org/10.1177/09596836231219482>
Code: <https://github.com/Archaeo-Programmer/paleoMAT>
-  2023 Kohler, Timothy A., Darcy Bird, **R. Kyle Bocinsky**, Kelsey Reese and Andrew D. Gillreath-Brown
Wealth inequality in the prehispanic northern US Southwest: from Malthus to Tyche. *Philosophical Transactions of the Royal Society B* 378: 1883.
DOI: <https://doi.org/10.1098/rstb.2022.0298>
Code: https://github.com/village-ecodynamics/kohler_et_al_2023_ptrsb
-  2022 Hoylman, Zachary Harwood, **R. Kyle Bocinsky**, and Kelsey G. Jencso
Drought assessment has been outpaced by climate change: Empirical arguments for a paradigm shift. *Nature Communications* 13: 2715.
DOI: <https://doi.org/10.1038/s41467-022-30316-5>
Code: <https://github.com/zhoylman/drought-year-sensitivity>
-  2022 Bird, Darcy, Lux Miranda, Marc Vander Linden, Erick Robinson, **R. Kyle Bocinsky**, Christopher Nicholson, José Capriles, Judson Finley, Eugenia Gayo, Adolfo Gil, Jade d'Alpoim Guedes, Julie Hoggarth, Andrea Kay, Emma Loftus, Umberto Lombardo, Madeline Mackie, Dr Alessio Palmisano, Dr Steinar Solheim, Robert Kelly, and Jacob Freeman
p3k14c, a synthetic global database of archaeological radiocarbon dates. *Scientific Data* 9: 27.
DOI: <https://doi.org/10.1038/s41597-022-01118-7>
Code: <https://github.com/people3k/p3k14c>
-  2021 Price, Michael Holton, José M. Capriles, Julie A. Hoggarth, **R. Kyle Bocinsky**, Claire E. Ebert, and James Holland Jones
End-to-end Bayesian analysis for summarizing sets of radiocarbon dates. *Journal of Archaeological Science* 135: 105473.
DOI: <https://doi.org/10.1016/j.jas.2021.105473>
Code: <https://github.com/eehh-stanford/price2020>
-  2021 Alam, Ornob, Rafal M. Gutaker, Cheng-chieh Wu, Karen A. Hicks, **R. Kyle Bocinsky**, Cristina Cobo Castillo, Stephen Acabado, Dorian Fuller, Jade A. d'Alpoim Guedes, Yue-ie Hsing, and Michael D. Purugganan
Genome analysis traces regional dispersal of rice in Taiwan and Southeast Asia. *Molecular Biology and Evolution*: msab209.
DOI: <https://doi.org/10.1093/molbev/msab209>
-  2021 Scheffer, Marten, Egbert H. van Nes, Darcy Bird, **R. Kyle Bocinsky**, and Timothy A. Kohler
Loss of resilience preceded transformations of pre-Hispanic Pueblo societies. *Proceedings of the National Academy of Sciences* 118(18): e2024397118.
DOI: <https://doi.org/10.1073/pnas.2024397118>
-  2021 Smith, Ada P., Laurie Yung, Adam Snitker, **R. Kyle Bocinsky**, Elizabeth Covelli Metcalf, and Kelsey Jencso
Scalar mismatches and underlying factors for underutilization of climate information: Perspectives from farmers and ranchers. *Frontiers in Climate* 3: 29 pages.
DOI: <https://doi.org/10.3389/fclim.2021.663071>

- 2021 Robinson, Erick, **R. Kyle Bocinsky**, Darcy Bird, Jacob Freeman, and Robert L. Kelly
Dendrochronological dates confirm a Late Prehistoric population decline in the American Southwest derived from radiocarbon dates. *Philosophical Transactions of the Royal Society B* 376: 20190718.
DOI: <https://doi.org/10.1098/rstb.2019.0718>
Code: <https://github.com/people3k/robinson2020>
- 2020 Strawhacker, Colleen, Grant Snitker, Matthew A. Peeples, Ann P. Kinzig, Keith W. Kintigh, **R. Kyle Bocinsky**, Brad Butterfield, Jacob Freeman, Sarah Oas, Margaret C. Nelson, Jonathan A. Sandor, Katherine A. Spielmann
A landscape perspective on climate-driven risks to food security: Exploring the relationship between climate and social transformations in the Prehispanic U.S. Southwest. *American Antiquity* 85(3): 427–451.
DOI: <https://doi.org/10.1017/aaq.2020.35>
Code: <https://github.com/gsnitker/LTVTP>
- 2020 Gutaker, Rafal M., Simon C. Groen, Emily S. Bellis, Jae Y. Choi, Inês S. Pires, **R. Kyle Bocinsky**, Emma R. Slayton, Olivia Wilkins, Cristina C. Castillo, Sónia Negrão, M. Margarida Oliveira, Dorian Q. Fuller, Jade A. d'Alpoim Guedes, Jesse R. Lasky, and Michael D. Purugganan
Genomic history and ecology of the geographic spread of rice. *Nature Plants* 6: 492–502.
DOI: <https://doi.org/10.1038/s41477-020-0659-6>
Code: <https://github.com/bocinsky/gutaker2020>
- 2019 Grimstead, Deanna N., Matthew C. Pailes, and **R. Kyle Bocinsky**
Refining potential source regions via combined maize niche modeling and isotopes: A case study from Chaco Canyon, NM, USA. *Journal of Archaeological Method and Theory* 26(1): 25–51.
DOI: <https://doi.org/10.1007/s10816-017-9359-6>
- 2018 Kohler, Timothy A., Philip I. Buckland, Keith W. Kintigh, **R. Kyle Bocinsky**, Adam Brin, Andrew Gillreath-Brown, Bertram Ludäscher, Timothy M. McPhillips, Rachel Opitz, and Jeffrey Terstriep
Paleodata for and from archaeology. *Past Global Changes Magazine* 26(2): 68–69.
DOI: <https://doi.org/10.22498/pages.26.2.68>
- 2018 d'Alpoim Guedes, Jade, and **R. Kyle Bocinsky**
Climate change stimulated agricultural innovation and exchange across Asia. *Science Advances* 4(10): eaar4491.
DOI: <https://doi.org/10.1126/sciadv.aar4491>
Code: <https://github.com/bocinsky/guedesbocinsky2018>
- 2017 d'Alpoim Guedes, Jade and **R. Kyle Bocinsky**
A reply to Bellezza. *Current Anthropology* 58(4): 539–539.
DOI: <https://doi.org/10.1086/692996>
- 2017 **Bocinsky, R. Kyle** and Mark D. Varien
Calibrating Maize Paleoproduction Models using Experimental Data. *Ethnobiology* 37(2): 282–307.
DOI: <https://doi.org/10.2993/0278-0771-37.2.282>
Code: https://github.com/bocinsky/Bocinsky_Varien_2017

- 2017 Gillreath-Brown, Andrew and **R. Kyle Bocinsky**
A Dialogue Between Empirical and Model-Based Agricultural Studies in Archaeology. *Ethnobiology* 37(2): 167–171.
DOI: <https://doi.org/10.2993/0278-0771-37.2.167>
Code: https://github.com/bocinsky/Bocinsky_Varien_2017
- 2017 Crabtree, Stefani A., **R. Kyle Bocinsky**, Paul L. Hooper, Susan C. Ryan, and Timothy A. Kohler
How to make a polity (in the central Mesa Verde region). *American Antiquity* 82(1): 71–95.
DOI: <https://doi.org/10.1017/aaq.2016.18>
- 2016 d’Alpoim Guedes, Jade, Stefani A. Crabtree, **R. Kyle Bocinsky**, and Timothy A. Kohler
21st century approaches to ancient problems: Climate and society. *Proceedings of the National Academy of Sciences* 113(51): 14483–14491.
DOI: <https://doi.org/10.1073/pnas.1616188113>
- 2016 d’Alpoim Guedes, Jade, Sturt Manning, and **R. Kyle Bocinsky**
A 5,500-year model of changing crop niches on the Tibetan Plateau. *Current Anthropology* 57(4): 6 pages.
DOI: <https://doi.org/10.1086/687255>
- 2016 Van Dyke, Ruth M., **R. Kyle Bocinsky**, Tucker Robinson, and Thomas C. Windes
Great houses, shrines, and high places: Intervisibility in the Chacoan World. *American Antiquity* 81(2): 205–230.
DOI: <https://doi.org/10.7183/0002-7316.81.2.205>
Code: https://github.com/bocinsky/VanDyke_et_al_2016
-  2016 **Bocinsky, R. Kyle**, Johnathan Rush, Keith W. Kintigh, and Timothy A. Kohler
Exploration and exploitation in the macrohistory of the pre-Hispanic Pueblo Southwest. *Science Advances* 2(4): e1501532.
DOI: <https://doi.org/10.1126/sciadv.1501532>
Code: https://github.com/bocinsky/Bocinsky_et_al_2016
- 2016 Schwindt, Dylan M., **R. Kyle Bocinsky**, Scott G. Ortman, Donna M. Glowacki, Mark D. Varien, and Timothy A. Kohler
The social consequences of climate change in the Central Mesa Verde region. *American Antiquity* 81(1): 74–96.
DOI: <https://doi.org/10.7183/0002-7316.81.1.74>
- 2016 Lipe, William D., **R. Kyle Bocinsky**, Brian S. Chisholm, Robin Lyle, David M. Dove, RG Matson, Elizabeth Jarvis, Kathleen Judd, and Brian M. Kemp
Cultural and genetic contexts for early turkey domestication in the northern Southwest. *American Antiquity* 81(1): 97–113.
DOI: <https://doi.org/10.7183/0002-7316.81.1.97>
-  2015 d’Alpoim Guedes, Jade, Guiyun Jin, and **R. Kyle Bocinsky**
The impact of climate on the spread of rice to North-eastern China: A new look at the data from Shandong Province. *PLoS ONE*. 30 June 2015.
DOI: <https://doi.org/10.1371/journal.pone.0130430>

-  2015 d'Alpoim Guedes, Jade, **R. Kyle Bocinsky**, and Ethan E. Butler
Comment on "Agriculture facilitated permanent occupation of the Tibetan Plateau after 3600 B.P." *Science* 348: 872.
DOI: <https://doi.org/10.1126/science.aaa4819>
-  2015 McPhillips, Timothy, Tianhong Song, Tyler Kolisni, Steve Aulenbach, Khalid Belhajjame, **R. Kyle Bocinsky**, Yang Cao, James Cheney, Fernando Chirigati, Saumen Dey, Juliana Freire, Christopher Jones, James Hanken, Keith W. Kintigh, Timothy A. Kohler, David Koop, James A. Macklin, Paolo Missier, Mark Schildhauer, Christopher Schwalm, Yaxing Wei, Mark Bieda, Bertram Ludäscher
YesWorkflow: A User-Oriented, Language-Independent Tool for Recovering Workflow Information from Scripts. *International Journal of Digital Curation* 10(1): 298–313.
DOI: <https://doi.org/10.2218/ijdc.v10i1.370>
- 2014 **Bocinsky, R. Kyle** and Timothy A. Kohler
A 2,000-year reconstruction of the rain-fed maize agricultural niche in the US Southwest. *Nature Communications* 5: 5618.
DOI: <https://doi.org/10.1038/ncomms6618>
- 2014 **Bocinsky, R. Kyle**
Extrinsic site defensibility and landscape-based archaeological inference: An example from the Northwest Coast. *Journal of Anthropological Archaeology* 35: 164–176.
DOI: <https://doi.org/10.1016/j.jaa.2014.05.003>
Code: https://github.com/bocinsky/Bocinsky_2014
-  2013 Cockburn, Denton, Stefani A. Crabtree, Ziad Kobti, Timothy A. Kohler, and **R. Kyle Bocinsky**
Simulating Social and Economic Specialization in Small-scale Agricultural Societies. *Journal of Artificial Societies and Social Simulation* 16(4): 4.
DOI: <https://doi.org/10.18564/jasss.2308>
- 2012 Kohler, Timothy A., **R. Kyle Bocinsky**, Denton Cockburn, Mark D. Varien, Stefani A. Crabtree, Kenneth E. Kolm, Schaun M. Smith, and Scott G. Ortman
Modelling Prehispanic Pueblo Societies in their Ecosystems. *Ecological Modelling* 241: 30–41.
DOI: <https://doi.org/10.1016/j.ecolmodel.2012.01.002>
- 2012 Kohler, Timothy A., Denton Cockburn, Paul L. Hooper, **R. Kyle Bocinsky** and Ziad Kobti
The coevolution of group size and leadership: An agent-based public goods model for Prehispanic Pueblo societies. *Advances in Complex Systems* 15(1–2): 29 pages.
DOI: <https://doi.org/10.1142/S0219525911003256>
- Book Contributions
* peer-reviewed
- * 2024 McCarthy, Maureen, Trent Teegerstrom, Kelsey Fitzgerald, Staci Emm, **Kyle Bocinsky**, and Vicki Hebb
Native Waters on Arid Lands, Southwestern U.S.: 2015–2022. In *World Drought Atlas*, pp. 108–109. Publications Office of the European Union.
DOI: <https://data.europa.eu/doi/10.2760/3842670>

- * 2023 Ermigiotti, Paul, Mark D. Varien, Grant D. Coffey, **R. Kyle Bocinsky**, Leigh Kuwanwisiwma, and Stewart B. Koyiyumptewa
The Pueblo Farming Project: Research, Education, and Native American Collaboration. In *Research, Education and American Indian Partnerships at the Crow Canyon Archaeological Center*, edited by Susan C. Ryan, Chapter 4, pp. 53–71. Routledge, New York.
DOI: <https://doi.org/10.5876/9781646424597>

- * 2023 Potter, James M., Mark D. Varien, Grant D. Coffey, and **R. Kyle Bocinsky**
Community Organization on the Edge of the Mesa Verde Region: Recent Investigations at Cowboy Wash Pueblo, Moqui Springs Pueblo, and Yucca House. In *Research, Education and American Indian Partnerships at the Crow Canyon Archaeological Center*, edited by Susan C. Ryan, Chapter 13, pp. 204–221. Routledge, New York.
DOI: <https://doi.org/10.5876/9781646424597>

- * 2023 Kohler, Timothy A., **R. Kyle Bocinsky**, and Darcy Bird
Fluctuat nec mergitur: Seven Centuries of Pueblo Crisis and Resilience. In *How Worlds Collapse: What History, Systems, and Complexity Can Teach Us About Our Modern World and Fragile Future*, edited by Miguel A. Centeno, Peter W. Callahan, Paul Larcey, and Thayer Patterson, Chapter 4, pp. 146–166. Routledge, New York.
DOI: <https://doi.org/10.4324/9781003331384-11>

- * 2021 Schachner, Gregson, Joel Nicholas, R. J. Sinensky, and **R. Kyle Bocinsky**
The sustainability of Hopi agriculture. In *Becoming Hopi: A History*, edited by Wesley Bernardini, Stewart Koyiyumptewa, Gregson Schachner, and Leigh Kuwanwisiwma, pp. 113–132. University of Arizona Press, Tucson.
HTML: <https://uapress.arizona.edu/book/becoming-hopi>
Winner of the 2022 Society for American Archaeology Scholarly Book Award.

- * 2021 Van Dyke, Ruth M., Timothy De Smet, and **R. Kyle Bocinsky**
Viewscapes and Soundscapes. In *The Greater Chaco Landscape: Ancestors, Scholarship, and Advocacy*, edited by Ruth Van Dyke and Carrie Heitman, Chapter 11. University Press of Colorado, Boulder.
DOI: <https://doi.org/10.5876/9781646421701>
Winner of the 2022 Society for American Archaeology Popular Book Award.
Winner of the 2021 American Anthropological Association Engaged Anthropology Award.

- * 2020 Kohler, Timothy A., Laura J. Ellyson and **R. Kyle Bocinsky**
Beyond One-Shot Hypotheses: Explaining Three Increasingly Large Collapses in the Northern Pueblo Southwest. In *Going forward by Looking Back: Archaeological Perspectives on Socio-Ecological Crisis, Response and Collapse*, edited by Felix Reide and Payson Sheets, pp. 304–332. Berghahn Books, New York.
DOI: <https://doi.org/10.3167/9781789208641>

- 2018 Ermigiotti, Paul, Mark Varien, Erin Bohm, **Kyle Bocinsky**, the Hopi Cultural Preservation Office, and the Hopi Cultural Resources Advisory Task Team
The Pueblo Farming Project: A collaboration between Hopi farmers and the Crow Canyon Archaeological Center. Crow Canyon Archaeological Center, Cortez, Colorado.
HTML: <https://pfp.crowcanyon.org>

- * 2018 Kohler, Timothy A., Stefani A. Crabtree, **R. Kyle Bocinsky**, and Paul L. Hooper
Sociopolitical Evolution in Midrange Societies: The Prehispanic Pueblo Case. In *The Emergence of Premodern States: New Perspectives on the Development of Complex Societies*, edited by Jeremy A. Sabloff and Paula L.W. Sabloff, pp. 133–186. SFI Press, Santa Fe, New Mexico.
HTML: <https://www.sfiypress.org/books/emergence-of-premodern-states>

- * 2017 Kohler, Timothy A. and **R. Kyle Bocinsky**
Crises as opportunities for culture change. In *Crisis to Collapse: The Archaeology of Social Breakdown*, edited by Tim Cunningham and Jan Driessen, pp. 263–273. Presses Universitaires de Louvain, Louvain, France.
HTML: <https://pul.uclouvain.be/book/?GCOI=29303100293170>

- * 2016 **Bocinsky, R. Kyle** and Timothy A. Kohler
Complexity, rigidity, and resilience in the Ancient Puebloan Southwest. In *Viewing the Future in the Past: Historical Ecology Applications to Environmental Issues*, edited by H. Thomas Foster, II, Lisa Paciulli, and David Goldstein. pp. 86–105. University of South Carolina Press, Columbia, South Carolina.
HTML: <https://uscpress.com/Viewing-the-Future-in-the-Past>

- * 2012 **Bocinsky, R. Kyle**, Jason A. Cowan, Timothy A. Kohler and C. David Johnson
Hunting results: How hunting changes the VEP world, and how the VEP world changes hunting. In *Emergence and Collapse of Early Villages: Models of Central Mesa Verde Archaeology*, edited by T.A. Kohler, and M.D. Varien, pp. 145–152. University of California Press, Berkeley.
HTML: <https://www.ucpress.edu/ebook/9780520951990/>

- * 2012 Kohler, Timothy A., **R. Kyle Bocinsky**, Stefani A. Crabtree and Ben Ford
Supply, Demand, Return Rates, and Resource Depression: Hunting in the Village Ecodynamics World. In *Emergence and Collapse of Early Villages: Models of Central Mesa Verde Archaeology*, edited by T.A. Kohler, and M.D. Varien, pp. 129–144. University of California Press, Berkeley.
HTML: <https://www.ucpress.edu/ebook/9780520951990/>

- * 2012 Cowan, Jason A., Timothy A. Kohler, C. David Johnson, Kevin Cooper and **R. Kyle Bocinsky**
Exercising the Model: Settlement Efficiency and Changes in Locational Practice. In *Emergence and Collapse of Early Villages: Models of Central Mesa Verde Archaeology*, edited by T.A. Kohler, and M.D. Varien, pp. 153–164. University of California Press, Berkeley.
HTML: <https://www.ucpress.edu/ebook/9780520951990/>

- * 2024 Hrozencik, Aaron, Gabriela Perez-Quesada, and **Kyle Bocinsky**
The Stocking Impact and Financial-Climate Risk of the Livestock Forage Disaster Program. Report No. ERR-329. U.S. Department of Agriculture, Economic Research Service.
HTML: <https://www.ers.usda.gov/publications/pub-details/?pubid=108371>

Technical Reports

* peer-reviewed

- 2023 **Bocinsky, R. Kyle**
Calculating normal grazing periods using historical nClimGrid-daily data. Montana Climate Office, Missoula, Montana.
Code: <https://github.com/mt-climate-office/nclimgrid-normal-grazing-period>
HTML: <https://mt-climate-office.github.io/nclimgrid-normal-grazing-period/>
- 2023 Fort Peck Assiniboiné and Sioux Tribes, Spackman Mossop Michaels, and the **Montana Climate Office**
Hazard Mitigation Plan for the Fort Peck Reservation, Montana. Fort Peck Assiniboiné and Sioux Tribes. August 2023 revision.
- 2020 **Bocinsky, R. Kyle**
Defining Accurate Floating Coordinate Reference Systems for Legacy Crow Canyon Projects. Crow Canyon Archaeological Center, Cortez, Colorado.
HTML: <https://crowcanyon.github.io/localgrid/articles/floating-grid-trigonometry.html>
- 2017 Glowacki, Donna M., **R. Kyle Bocinsky**, Kelsey M. Reese, and Katherine A. Portman
The Mesa Verde Community Center Survey: Summer 2017. Manuscript submitted to Mesa Verde National Park, on file with Crow Canyon Archaeological Center and the University of Notre Dame.
- 2012 Glowacki, Donna M., with contributions by Elise Alonzi, **R. Kyle Bocinsky**, and Kelsey Reese
The Mesa Verde Community Center Survey: Final Report. Manuscript submitted to the National Science Foundation and Mesa Verde National Park, on file with Crow Canyon Archaeological Center and Washington State University in compliance with deliverable for NSF Grant DEB-0816400.
- 2011 Glowacki, Donna M. and **R. Kyle Bocinsky**
The Mesa Verde Community Center Survey: 2011 Progress Report. Manuscript submitted to the National Science Foundation and Mesa Verde National Park, on file with Crow Canyon Archaeological Center and Washington State University in compliance with deliverable for NSF Grant DEB-0816400.
- 2009 Reed, Charles A., **R. Kyle Bocinsky** and Donna M. Glowacki
Mesa Verde National Park Community Center Survey: Report and Synthesis. Manuscript submitted to the National Science Foundation and Mesa Verde National Park, on file with Crow Canyon Archaeological Center and Washington State University in compliance with deliverable for NSF Grant DEB-0816400.

Other Publications

- Press Kohler, Tim, **Kyle Bocinsky**, Mark Varien, and Scott Ortman
The Village Ecodynamics Project: A Recollection. *Archaeology Southwest*.
HTML: <https://www.archaeologysouthwest.org/product/asw32-3/>
- 2020 **Bocinsky, R. Kyle**
Review of *The Archaic Southwest: Foragers in an Arid Land*, edited by Bradley J. Vierra. *Journal of California and Great Basin Anthropology* 40(2): 244–245.
HTML: http://malkimuseum.org/wp-content/uploads/2017/02/jcgba_40-2_toc.pdf

- 2018 **Bocinsky, R. Kyle**
Climate at the Center Place. *Archaeology Southwest* 32(2–3): 32–33.
HTML: <https://www.archaeologysouthwest.org/product/asw32-3/>
- 2018 **Bocinsky, R. Kyle**
Skipping stones off improbable histories. *The SAA Archaeological Record* 18(2): 17–19.
HTML: <http://onlinedigeditions.com/publication/?m=16146&i=486133&p=19>
- 2018 Marwick, Ben, Jade d'Alpoim Guedes, C Michael Barton, Lynsey A Bates, Michael Baxter, Andrew Bevan, Elizabeth A Bollwerk, **R. Kyle Bocinsky**, Tom Brughmans, Alison K Carter, Cyler Conrad, Daniel A Contreras, Stefano Costa, Enrico R Crema, Adrianne Daggett, Benjamin Davies, B Lee Drake, Thomas S Dye, Phoebe France, Richard Fullagar, Domenico Giusti, Shawn Graham, Matthew D Harris, John Hawks, Damien Huffer, Eric C Kansa, Sarah Whitcher Kansa, Mark E Madsen, Jennifer Melcher, Joan Negre, Fraser D Neiman, Rachel Opitz, David C Orton, Paulina Przstupa, Maria Raviele, Julien Riel-Savatore, Philip Riris, Iza Romanowska, Jolene Smith, Nehemie Strupler, Isaac I Ullah, Hannah G Van Vlack, Nathaniel VanValkenburgh, Ethan C Watrall, Chris Webster, Joshua Wells, Judith Winters, and Colin D Wren
Open science in archaeology. *The SAA Archaeological Record* 17(4): 8–14.
HTML: <http://onlinedigeditions.com/publication/?m=16146&i=440506&p=10>
- 2014 **Bocinsky, R. Kyle**
Drain your dammed DEMs! Here's how. *Society for Archaeological Sciences Bulletin* 37(1): 1–5.
HTML: <https://socarchsci.org/bulletin/SAS3701.pdf>
- Software**
- *^c 2025 **Bocinsky, R. Kyle**
FedData: Functions to Automate Downloading Geospatial Data Available from Several Federated Data Sources. R package version 4.3.0
URL: <https://github.com/bocinsky/FedData>
- *^c 2023 Bevan, Andrew, Enrico Crema, **R. Kyle Bocinsky**, Martin Hinz, Philip Riris, and Fabio Silva
rcarbon: Calibration and Analysis of Radiocarbon Dates. R package version 1.5.1
URL: <https://github.com/ahb108/rcarbon>
- 2020 **Bocinsky, R. Kyle**
thredds: Access to THREDDS Servers. R package version 0.0.1
URL: <https://github.com/bocinsky/thredds>
- 2022 **Bocinsky, R. Kyle**
mcor: The core Montana Climate Office R package. R package version 0.2.0
URL: <https://github.com/mt-climate-office/mcor>
- * 2021 Price, Michael Holton and **R. Kyle Bocinsky**
baydem: Bayesian Tools for Reconstructing Past and Present Demography. R package version 1.0.0
URL: <https://github.com/eehh-stanford/baydem>
- 2019 **Bocinsky, R. Kyle**
demdrainer: Drain Your Dammed DEMs. R package version 0.1.0
URL: <https://github.com/bocinsky/demdrainer>

- 2019 **Bocinsky, R. Kyle**
tdar: R Bindings to the Digital Archaeological Record (tDAR) API. R package version 0.1.0
 URL: <https://github.com/bocinsky/tdar>
- 2019 **Bocinsky, R. Kyle**
treecats: Tree-like Factors in R. R package version 0.1.0
 URL: <https://github.com/bocinsky/treecats>
- * 2021 **Bocinsky, R. Kyle**
paleocar: Paleoclimate Reconstruction from Tree Rings using Correlation Adjusted corRe-lation. R package version 3.0.0
 URL: <https://github.com/bocinsky/paleocar>

Conference Presentations

- 2025 **Bocinsky, R. Kyle** and Ian McRyhew
 Healing Stories: The Native Resilience Drought Projects. Drought and Health National Workshop, Science Museum of Minnesota, Saint Paul, Minnesota. September 30–October 1, 2025.
- 2025 **Bocinsky, R. Kyle**, Shamsi Daneshvari Berry, Jeffery Clark, Keith Kintigh, and Scott Ortman
 Developing a Research Loom for Weaving Stories of Sustainability. In symposium, *Multiscale Data and the History of Human Development in the US Southwest*. Society for American Archaeology, 90th Annual Meeting, Denver, Colorado.
- 2025 Davis, Kaitlyn, and **R. Kyle Bocinsky**
 More Than Maize: Modeling the Cotton, Wheat, (and Maize) Cultivation Niches beyond the Four Corners. In symposium, *Multiscale Data and the History of Human Development in the US Southwest*. Society for American Archaeology, 90th Annual Meeting, Denver, Colorado.
- 2025 Barbara Mills, **R. Kyle Bocinsky**, Jeffery Clark, Kaitlyn Elizabeth Davis, and Sarah Oas
 Cotton Production and Regional Distribution for Western Pueblo Cultural and Ritual Sustainability, 1150–1450 CE. In symposium, *Multiscale Data and the History of Human Development in the US Southwest*. Society for American Archaeology, 90th Annual Meeting, Denver, Colorado.
- 2025 Ann Stodder, Lexi O'Donnell, Catrina Whitley, **R. Kyle Bocinsky**, and Kenneth Vernon
 Through the Biocultural Lens: Resilience, Vulnerability, and Lived Experience in the Ancient Southwest. In symposium, *Multiscale Data and the History of Human Development in the US Southwest*. Society for American Archaeology, 90th Annual Meeting, Denver, Colorado.
- 2024 **Bocinsky, R. Kyle** and Maureen I. McCarthy
 All Climate is Local: Droughts to Floods on Native Lands. In symposium, *Supporting Tribal Drought Resilience*. 2024 National Tribal and Indigenous Climate Conference, Anchorage, Alaska. September 10, 2024.
- 2023 **Bocinsky, R. Kyle**, Maureen I. McCarthy, Meghan Collins, Staci Emm, Kelsey Fitzgerald, Vicki E. Hebb, Kelsey Jencso, Ileah Kirchoff, and Trent Teegerstrom
 Strengthening the Role of Climate Hubs in Indian Country: Introducing the Native Climate Project. In symposium, *Translating Climate Science into Action with the USDA Climate Hubs*. 103rd American Meteorological Society Annual Meeting, Denver, Colorado. January 10, 2023.

- 2023 Jencso, Kelsey, Zachary H. Hoylman, **R. Kyle Bocinsky**, Kevin Hyde, Joel Martin, and Colin Brust
Navigating the Waters Ahead: The Development of a Climate Monitoring and Drought Assessment Framework for Montana. In symposium, *Advancements in Analysis and Prediction of Drought II*. 103rd American Meteorological Society Annual Meeting, Denver, Colorado. January 12, 2023.
- 2022 Plenty Sweetgrass-She Kills, Ruth, Jennifer Harrington, Kyle Bocinsky, Amber Finley, Judy Dudley, John Phillips, Marea DeVille, Gerri Pearson, Selena Finley, and Paige Johnson
CU Missouri River Basin Consortium. 2022 First American Land Grants Consortium (FALCON) Annual Conference, Albuquerque, New Mexico. October 20, 2022.
- 2022 **Bocinsky, R. Kyle**
Who Owns Tribal Climate Data? A micro-workshop on Climate Data Sovereignty. Second National Tribal & Indigenous Climate Conference, St. Paul, Minnesota. August 29, 2022.
- 2022 Bird, Darcy, Andrew Gillreath-Brown, **R. Kyle Bocinsky**, and Tim Kohler
Climate, Maize, and Demography: A Case Study from the Southwestern United States. In symposium, *Climatic Influences on Diet, Demography, and Conflict*. Society for American Archaeology, 87th Annual Meeting, Chicago, Illinois.
- 2022 Kohler, Tim, Andrew Gillreath-Brown, **R. Kyle Bocinsky**, and Darcy Bird
From Unconstrained to Density-dependent Growth: 900 years of Pueblo pre/history. In symposium, *Demography in Archaeology beyond Boom and Bust? Critical Examination of Phases of Under- and Overpopulation*. European Association of Archaeologists, 27th Annual Meeting, Kiel, Germany.
- 2021 Bird, Darcy, **R. Kyle Bocinsky**, and Tim Kohler
Maize, Construction, and Population Changes: One Way to Identify Sunk Cost Behaviors in Central Mesa Verde. In symposium, *People, Climate, and Proxies in Holocene Western North America*. Society for American Archaeology, 86th Annual Meeting, online.
- 2019 Erick Robinson, Jacob Freeman, Robert L. Kelly, and **R. Kyle Bocinsky**
The Role of Edge Effects in Late Holocene Archaeological Radiocarbon Time Series. In symposium, *Global Perspectives on Climate-Human Population Dynamics During the Late Holocene*. Society for American Archaeology, 84th Annual Meeting, Albuquerque, New Mexico.
- 2019 Gillreath-Brown, Andrew, **R. Kyle Bocinsky**, and Tim Kohler
The Impact of Temperature on the Transition to Maize Agriculture in the Northern Upland United States Southwest. In symposium, *Global Perspectives on Climate-Human Population Dynamics During the Late Holocene*. Society for American Archaeology, 84th Annual Meeting, Albuquerque, New Mexico.
- 2019 Glowacki, Donna, Mark Varien, Grant Coffey and **R. Kyle Bocinsky**
Mesa Verde Centers and Regional Analyses: Good Stuff!. In symposium, *Attention to Detail: A Pragmatic Career of Research, Mentoring, and Service, Papers in Honor of Keith Kintigh*. Society for American Archaeology, 84th Annual Meeting, Albuquerque, New Mexico.
- 2019 Coffey, Grant, Mark Varien and **R. Kyle Bocinsky**
Basketmaker III in the Central Mesa Verde Region: Transitions, Social Dynamics, and Population Growth. In symposium, *Adopting the Pueblo Fettle: The Breadth and Depth of the Basketmaker III Cultural Horizon*. Society for American Archaeology, 84th Annual Meeting, Albuquerque, New Mexico.

- 2019 **Bocinsky, R. Kyle**, Andrew Gillreath-Brown and Tim Kohler
The Climates of Pueblo Emergence. In symposium, *Adopting the Pueblo Fettle: The Breadth and Depth of the Basketmaker III Cultural Horizon*. Society for American Archaeology, 84th Annual Meeting, Albuquerque, New Mexico.
- 2018 **Bocinsky, R. Kyle**
Directing the arrow by chasing its tail: Climate, adaptation, and future (agri)cultural systems in the West. In plenary symposium, *Arrow of Time: Anthropological Disciplines of the Future*. 36th Great Basin Anthropological Conference, Salt Lake City, Utah.
- 2018 Glowacki, Donna and **R. Kyle Bocinsky**
The Great Houses of the Mesa Verde Cuesta. In symposium, *Mesa Verde Ancestral Pueblo Villages: Recent Research*. Society for American Archaeology, 83rd Annual Meeting, Washington, DC.
- 2018 Gillreath-Brown, Andrew, **R. Kyle Bocinsky**, Simon Goring and Tim A. Kohler
Paleotemperature Reconstructions of the Upland United States Southwest for the Last 2,000 Years. In general session, *Environmental Archaeology*. Society for American Archaeology, 83rd Annual Meeting, Washington, DC.
- 2017 d'Alpoim Guedes, Jade, and **R. Kyle Bocinsky**
Climate Change, Agricultural Innovation and Exchange across Asia. American Geophysical Union, Fall Meeting 2017, New Orleans, Louisiana.
- 2017 **Bocinsky, R. Kyle**
"Constraint and Freedom" in the Era of Big Data. In Frison Institute Symposium, *The Future of "Big Data" in Archaeology*. Society for American Archaeology, 82nd Annual Meeting, Vancouver, British Columbia, Canada.
- 2017 Lipe, William, Laura Ellyson, **R. Kyle Bocinsky**, Robin Lyle and R.G. Matson
Costly Gobbling: Raising Turkeys in the Central Mesa Verde Area. In general session, *The Archaeology of Foodways*. Society for American Archaeology, 82nd Annual Meeting, Vancouver, British Columbia, Canada.
- 2017 d'Alpoim Guedes, Jade, and **R. Kyle Bocinsky**
Modeling the Spread of Crops across Eurasia. In Fryxell Award Symposium, *Papers in Honor of Naomi F. Miller*. Society for American Archaeology, 82nd Annual Meeting, Vancouver, British Columbia, Canada.
- 2016 Stodden, Victoria, Bertram Ludaescher, **Kyle Bocinsky**, Keith Kintigh, Tim Kohler, Timothy McPhillips, and Johnathan Rush
Querying Provenance Information: Basic Notions and an Example from Paleoclimate Reconstruction. American Geophysical Union, Fall Meeting 2016, San Francisco, California.
- 2016 Varien, Mark. D and **R. Kyle Bocinsky**
The Pueblo Farming Project: Investigating the Agrarian Ecology of the Mesa Verde Region. In symposium, *First Farmers, First Farms: Landscape Ecology of the Early Neolithic*. Society of Ethnobiology, 39th Annual Conference, Tucson, Arizona.
- 2016 **Bocinsky, R. Kyle** and Adam Brin
SKOPE: Bringing Continent-scale, Local Paleoenvironmental Data to Researchers and the Public. In symposium, *What Do We Mean by "Digital Curation?"* Society for American Archaeology, 81st Annual Meeting, Orlando, Florida.

- 2016 d'Alpoim Guedes, Jade, Sturt Manning, and **R. Kyle Bocinsky**
5500 Years of Changing Crop Niches on the Tibetan Plateau. In symposium, *Guava Cat Archaeology: Papers in Memory of Prof. Pochan Chen*. Society for American Archaeology, 81st Annual Meeting, Orlando, Florida.
- 2016 Ellyson, Laura, Timothy Kohler, and **R. Kyle Bocinsky**
Evaluation of the Village Ecodynamics Hunting and Domestication Models. In general session, *Southwestern Archaeology*. Society for American Archaeology, 81st Annual Meeting, Orlando, Florida.
- 2016 Van Dyke, Ruth M., **R. Kyle Bocinsky**, Tucker Robinson, and Thomas C. Windes
Great houses, shrines, and high places: Intervisibility across the Chacoan World. 2016 Navajo Nation Archaeology Meeting, Shiprock, New Mexico. 4 March 2016.
- 2016 **Bocinsky, R. Kyle**, Jade d'Alpoim Guedes, Karen Adams, Rob Quinlan, Mark Caudell, and Timothy A. Kohler
Can Pueblo Corn Save Ethiopian Farms? Deploying 1,400 Years of Agricultural Knowledge in Service of the Future. In symposium, *Engaged Archaeology through Transnational, Interdisciplinary, and Indigenous Collaborations*. 15th Biennial Southwest Symposium, Tucson, Arizona.
- 2016 Schwindt, Dylan M., **R. Kyle Bocinsky**, Scott G. Ortman, Donna M. Glowacki, Mark D. Varien, and Timothy A. Kohler
Interactions among society, climate, production, and population: A case study from the central Mesa Verde region with contemporary implications. In symposium, *Engaged Archaeology through Transnational, Interdisciplinary, and Indigenous Collaborations*. 15th Biennial Southwest Symposium, Tucson, Arizona.
- 2015 **Bocinsky, R. Kyle**
What is the Pecos System, Anyway? 2015 Pecos Conference, Mancos, Colorado.
Winner of the 2015 Cordell Prize.
- 2015 **Bocinsky, R. Kyle**, Keith Kintigh, Timothy Kohler, Bertram Ludäscher, Ann Kinzig, Carla Van West, Margaret Nelson, Adam Brin, Sarah Oas, Johnathan Rush, Timothy McPhillips, and Yang Cao
Toward Effective Cyber-Infrastructure Support of Socio-Environmental Research. In symposium, *The Robustness and Vulnerability of Food Production and Social Change: An Evaluation of Interdisciplinary Concepts Using Archaeological Data, Models, and Ethnographic Observations*. Society for American Archaeology, 80th Annual Meeting, San Francisco, California.
- 2015 Van Dyke, Ruth, **R. Kyle Bocinsky**, and Tucker Robinson
Friends in High Places: Windes, Shrines, and Lines of Sight. In symposium, *It's About Time: Contributions in Honor of Thomas C. Windes*. Society for American Archaeology, 80th Annual Meeting, San Francisco, California.
- 2015 Kohler, Timothy A., Stefani A. Crabtree, and **R. Kyle Bocinsky**
The Effects of Temporal Coarse-Graining on Inferred Networks of Human Movement. In symposium, *The Short and Long of It: Combining Timescales*. Society for American Archaeology, 80th Annual Meeting, San Francisco, California.
- 2015 Crabtree, Stefani A., **R. Kyle Bocinsky**, and Timothy A. Kohler
Alliances, Coalitions, Hierarchies and Conflict in the Ancestral Pueblo World. In symposium, *Simulating Social Complexity to Understand the Archaeological Past*. Society for American Archaeology, 80th Annual Meeting, San Francisco, California.

- 2015 McPhillips, Timothy, Tianhong Song, Tyler Kolisnik, Steve Aulenbach, Khalid Belhajjame, **R. Kyle Bocinsky**, Yang Cao, Fernando Chiriga, Saumen Dey, Juliana Freire, Deborah Huntzinger, Christopher Jones, David Koop, Paolo Missier, Mark Schildhauer, Christopher Schwalm, Yaxing Wei, James Cheney, Mark Bieda, and Bertram Ludäscher
YesWorkflow: A User-Oriented, Language-Independent Tool for Recovering Workflow Information from Scripts. 10th International Digital Curation Conference, London.
- 2014 Crabtree, Stefani A., **R. Kyle Bocinsky**, and Timothy A. Kohler
Modeling Emergence of Complex Groups on Realistic Landscapes. In symposium, *Principles of Complexity: Life, Scale, and Civilization III*. Santa Fe Institute, Santa Fe, New Mexico. August.
- 2014 **Bocinsky, R. Kyle**, Kurt F. Anschultz, Jesse Clark, and Timothy A. Kohler
A General Spatial Reconstruction of Potential Maize Paleoproductivity, AD 600–2000. In symposium, *Coupled Regions, Coupled Systems: Dynamics of Prehispanic Farming Societies in the Northern San Juan and the Northern Rio Grande*. Society for American Archaeology, 79th Annual Meeting, Austin, Texas.
- 2014 Kohler, Timothy A., Kelsey M. Reese, Stefani A. Crabtree, **R. Kyle Bocinsky**, and Brian M. Kemp
Prolegomenon: VEP II, Almost in Retrospect. In symposium, *Coupled Regions, Coupled Systems: Dynamics of Prehispanic Farming Societies in the Northern San Juan and the Northern Rio Grande*. Society for American Archaeology, 79th Annual Meeting, Austin, Texas.
- 2014 Crabtree, Stefani A., Timothy A. Kohler and **R. Kyle Bocinsky**
The Development of Social Groups, Leadership and Inequality in the Central Mesa Verde. In symposium, *Coupled Regions, Coupled Systems: Dynamics of Prehispanic Farming Societies in the Northern San Juan and the Northern Rio Grande*. Society for American Archaeology, 79th Annual Meeting, Austin, Texas.
- 2014 Conway, Meagan and **R. Kyle Bocinsky**
Household and Community Scales of Post-Famine Demographic Change in Western Ireland. In symposium, *The Archaeology of the Irish Famine: Exploring Lasting Effects on Local, Regional, and Transatlantic Scales*. Society for American Archaeology, 79th Annual Meeting, Austin, Texas.
- 2013 **Bocinsky, R. Kyle**
Pueblo II and the Turkey Taboo: Changing Attitudes Towards Domesticated Turkeys in the Puebloan Southwest. In symposium, *Food: It's What's for Dinner*. American Anthropological Association, 2013 Annual Meeting, Chicago, Illinois.
- 2013 Kohler, Timothy A., **R. Kyle Bocinsky**, Stefani A. Crabtree, Andrew Hlynka, Lokesh Patil, Paul Lovell Hooper and Ziad Kobti
Leadership, Territoriality and Conflict in an Agentized Pueblo World. In symposium, *Complex-systems Approaches to Long-term Change*. American Anthropological Association, 2013 Annual Meeting, Chicago, Illinois.
- 2013 **Bocinsky, R. Kyle**
"Free nights and weekends" in the AD 1000s? Prehistoric visual communication systems and the influence of infrastructure on modern social network formation. In lecture series, *Reckless Ideas*. Santa Fe Institute, Santa Fe, New Mexico. August 23, 2013.

- 2013 **Bocinsky, R. Kyle**
The Data Are the Context: A Student's Perspective on the Colorado Contexts, Old and New. In symposium, *Updating the 1999 Colorado Prehistoric Contexts: How a Digital Research Context for Southwestern Colorado Could Be Organized and Assembled*. Colorado Council of Professional Archaeologists, 35th Annual Conference, Denver, Colorado.
- 2013 **Bocinsky, R. Kyle**
The Defensive Coast. In general session, *Landscapes and Spatial Analysis: Global Case Studies*. Society for American Archaeology, 78th Annual Meeting, Honolulu, Hawaii.
Winner of the 2013 R.E. Taylor Student Poster Award from the Society for Archaeological Sciences.
- 2013 Reese, Kelsey M., **R. Kyle Bocinsky**, Donna M. Glowacki, Scott G. Ortman and Timothy A. Kohler
What Are Communities, Really? In symposium, *New Directions in Modeling Dynamics for Coupled Social-Natural Systems*. Society for American Archaeology, 78th Annual Meeting, Honolulu, Hawaii.
- 2013 **Bocinsky, R. Kyle**
Why Landscapes Matter: Inferring Defensiveness around the Salish Sea. In symposium, *Why Salish Sea Archaeology Matters: Current Perspectives from Washington State University*. 66th Annual Northwest Anthropological Conference, Portland, Oregon.
- 2012 **Bocinsky, R. Kyle**
Simulating Archaeological Landscapes using the RePAST Agent Based Modeling Framework. In invited colloquium, Institute of Archaeology and Museology, Faculty of Arts, Masaryk University, Brno, Czech Republic.
- 2012 **Bocinsky, R. Kyle** and Timothy A. Kohler
Suboptimal Foraging Theory (It's the Best). In symposium, *Socionatural Systems in the Northern U.S. Southwest: A Village Ecodynamics Project II Progress Report*. Society for American Archaeology, 77th Annual Meeting, Memphis, Tennessee.
- 2012 Kohler, Timothy A. and **R. Kyle Bocinsky**
Highly Variable Temperatures in the Northern Hemisphere over the Last Two Millennia: Reconstructions and Possible Effects on Native American Agricultural Systems. In symposium, *What Role did Changing Temperatures Play in the Success of North American Farming Societies in the Last Prehispanic Millennium?* Society for American Archaeology, 77th Annual Meeting, Memphis, Tennessee.
- 2012 Newbold, Bradley, **R. Kyle Bocinsky**, William D. Lipe, Phil R. Geib and Brian M. Kemp
Revising Early Turkey Domestication in the Southwest. In general session, *Archaeology of the Ancestral Puebloans and Paquime*. Society for American Archaeology, 77th Annual Meeting, Memphis, Tennessee.
- 2012 Reese, Kelsey M., **R. Kyle Bocinsky**, Kay E. Barnett, Stefani A. Crabtree and Elise Alonzi
Large, Aggregated Villages in Little Soda Canyon. In general session, *Southwest*. Society for American Archaeology, 77th Annual Meeting, Memphis, Tennessee.
- 2011 **Bocinsky, R. Kyle** and Timothy A. Kohler
Toward [Re]Constructing An Archaeological Legacy of Hunting On the Colorado Plateau. In Evolutionary Anthropology Society invited symposium, *Lasting Legacies and Changing Tides in Human Behavioral Ecology*. American Anthropological Association, 2011 Annual Meeting, Montreal, Quebec, Canada.

- 2011 **Bocinsky, R. Kyle**, Brian Chisholm and Brian M. Kemp
Basketmaker III Turkey Use: Multiple Lines of Evidence. In symposium, *The Cedar Mesa Project Turns 40: New Results from a Long-Lived Study in SE Utah*. Society for American Archaeology, 76th Annual Meeting, Sacramento, California.
- 2011 Newbold, Bradley, **R. Kyle Bocinsky**, Brian M. Kemp and Phil R. Geib
Ancient Domestic Turkeys at Atlatl Rock Cave. In general session, *Animal Domesticates in the Americas*. Society for American Archaeology, 76th Annual Meeting, Sacramento, California.
- 2011 Crabtree, Stefani A., Ziad Kobti, Timothy A. Kohler and **R. Kyle Bocinsky**
Archaeology Time Machine: Witnessing Social Interactions on the Pueblo Landscape through Agent-based Simulation. In symposium, *Mapping and Technological Experiments in Archaeology*. Society for American Archaeology, 76th Annual Meeting, Sacramento, California.
- 2011 Kohler, Timothy A., **R. Kyle Bocinsky**, Stefani A. Crabtree and Denton Cockburn
The Village Ecodynamics Project: Modeling the Deep Pueblo Past. In symposium, *Modeling Across Millennia: Interdisciplinary Paths to Ancient Socionatural Systems*. 177th AAAS Annual Meeting, Washington, D.C.
- 2011 Barron, Doug, **R. Kyle Bocinsky**, Dan Grunspan, Bradley Newbold, Darryl Trumbo and Bryan Roosien
Mapping the Memetics Meme: Toward a better understanding of the transmission and evolution of scientific concepts. In IGERT Program in Evolutionary Modeling seminar, Washington State University, Pullman, Washington.
- 2010 **Bocinsky, R. Kyle**, Breanne M. Nott and Brian M. Kemp
Plop! Jump-starting the brown revolution in archaeology. In symposium, *The Village Ecodynamics Project II*. Society for American Archaeology, 75th Annual Meeting, St. Louis, Missouri.
- 2010 Glowacki, Donna M., Charles A. Reed, **R. Kyle Bocinsky**, Shanna Diederichs and Julie Bell
Making sense of the actual: Settlement trends in the southwestern Colorado VEP study areas. In symposium, *The Village Ecodynamics Project II*. Society for American Archaeology, 75th Annual Meeting, St. Louis, Missouri.
- 2010 Reed, Charles A., Ryan Lash and **R. Kyle Bocinsky**
An Ancestral Puebloan Community in Morefield Canyon, Mesa Verde National Park. In general session, *What's New in Southwestern Archaeology?* Society for American Archaeology, 75th Annual Meeting, St. Louis, Missouri.
- 2009 **Bocinsky, R. Kyle**
In the land of plenty: Agent-based approaches to modeling subsistence in the Puebloan Southwest. In South Carolina Institute of Archaeology and Anthropology seminar, *From Field to Table: Historical Ecology of Regional Subsistence Strategies*. University of South Carolina, Columbia, South Carolina.
- 2009 **Bocinsky, R. Kyle**
Village Evolving: New directions in VEP research. In NSF Office of Polar Programs: Arctic Social Sciences sponsored conference, *Global Long Term Human Ecodynamics*. Eagle Hill, Maine.
- 2009 **Bocinsky, R. Kyle**
Understanding and modeling turkey use in the American Southwest: A preliminary simulation module for RePAST. In general session, *General New World and Methods*. Society for American Archaeology, 74th Annual Meeting, Atlanta, Georgia.

Forums, Workshops, & Symposia

- 2008 **Bocinsky, R. Kyle**, Terrence Martin and Mark A. Schurr
Rodent Stable Carbon-isotope Ratios as a Measure of Maize Production. In general session, *Methods and Advances in Zooarchaeology*. Society for American Archaeology, 73rd Annual Meeting, Vancouver, British Columbia, Canada.
- 2025 Co-Organizer & Discussant, *Thinking of Acronyms: A Kohler Obsession? Papers in Honor of Timothy A. Kohler (TAKO)*. Symposium, Society for American Archaeology, 90th Annual Meeting, Denver, Colorado.
- 2025 Panelist, *Rural Community Climate Resilience Series*. Workshop hosted by 350 Montana and the Montana Health Professionals for a Healthy Climate. Poulson, Libby, and Thompson Falls, Montana. September 26–30, 2025.
- 2025 Participant, *Drought and Health National Workshop*, Science Museum of Minnesota, Saint Paul, Minnesota. September 30–October 1, 2025.
- 2024 Co-Organizer & working group facilitator, *Climate Resilience Through Storytelling*. Pre-conference workshop hosted by the Native Resilience Project. 20th Annual Conference of the First Americans Land-Grant Consortium (FALCON), Minneapolis, Minnesota. October 11–13, 2024
- 2024 Co-Organizer & working group facilitator, *Indigenous ground truths: What is your climate story?*. Workshop hosted by the Native Resilience Project. 2024 National Tribal and Indigenous Climate Conference, Anchorage, Alaska. September 9, 2024.
- 2024 Co-Organizer & working group facilitator, *Bridging Waters: A Collaborative Approach to Addressing Water Quality, Water Scarcity, and Climate Change*. Workshop Co-hosted by the Native Resilience Project and University of Alaska, Anchorage. 2024 National Tribal and Indigenous Climate Conference, Anchorage, Alaska. September 9, 2024.
- 2023 Planning team & working group facilitator, *Workshop for Building Drought Resilience in a Changing Climate with Upper Columbia and Missouri Basin Tribes*. Workshop Co-hosted by NOAA NIDIS and the Confederated Salish and Kootenai Tribes, Polson, Montana. September 26–28, 2023.
- 2022 Co-Organizer, *Maize Database Project All-hands Meeting with Hopi Cultural Preservation Office*. Workshop, Flagstaff, Arizona. September 7–10, 2022.
- 2022 Co-Organizer, *All Climate Is Local: A resilience workshop for TCU extension, faculty, and students*. Workshop, 2022 First American Land Grants Consortium Annual Conference, Albuquerque, New Mexico.
- 2019 Participant, *Healthy Forests, Healthy Watersheds*, Workshop, Laboratory of Tree Ring Research, University of Arizona, Tucson, Arizona. November 13–15, 2019.
- 2019 Organizer, *Plenary Workshop – Native Waters on Arid Lands: Using wisdom from the past to guide the future of traditional practices, farming, ranching, forestry, and natural resource management*. Workshop, Fifth Annual Native Waters on Arid Lands Tribal Summit, Reno, Nevada.
- 2018 Organizer, *Building Research Capacity at Tribal Colleges and Universities and in Tribal Communities*. Forum, Fourth Annual Native Waters on Arid Lands Tribal Summit, Reno, Nevada.
- 2018 Discussant, *SKOPE: How we are making paleoenvironmental data useful and accessible*. Forum, Society for American Archaeology, 83rd Annual Meeting, Washington, DC.

- 2018 Participant, *Conceptualizing a Geospatial Software Institute (GSI)*. Workshop 2: Geospatial Use Cases and Core Technical Capabilities, CyberGIS Center for Advanced Digital and Spatial Studies, Chicago, Illinois.
- 2017 Guest editor, *First Farmers, First Farms: Landscape Ecology of the Early Neolithic*. Edited Section, Journal of Ethnobiology.
- 2017 Organizer, *Traditional Knowledge & Ecology—Building Agri-Cultural Resilience through Traditional Values*. Forum, Third Annual Native Waters on Arid Lands Tribal Summit, Reno, Nevada.
- 2017 Co-organizer, *Connecting Mesa Verde and the Rio Grande through the Study of Pueblo Foodways*. Forum, New Mexico History Museum, Santa Fe, New Mexico. 22 April 2017.
- 2017 Discussant, *Aztec at the Crossroads and in the Crosshairs: 101 Years of Research in One Convenient Symposium*. Forum, Society for American Archaeology, 82nd Annual Meeting, Vancouver, British Columbia, Canada.
- 2017 Discussant, *How to do Archaeological Science using R*. Forum, Society for American Archaeology, 82nd Annual Meeting, Vancouver, British Columbia, Canada.
- 2017 Participant, *Whole Tale: Computing environments for reproducibility*. Prov-a-Thon and architecture workshop, Santa Ana Pueblo, New Mexico.
- 2016 Discussant, *Traditional Knowledge & Ecology*. Forum, Second Annual Native Waters on Arid Lands Tribal Summit, Las Vegas, Nevada.
- 2016 Organizer, *Archaeologies of Inclusion: Towards a Politics of Engagement*. Forum, Society for American Archaeology, 81st Annual Meeting, Orlando, Florida.
- 2015 Organizer, *Out in the Field: Queer Experiences and Challenges in Archaeology*. Forum, Society for American Archaeology, 80th Annual Meeting, San Francisco, California.
- 2014 Discussant, *Error, Sensitivity Analysis, and Uncertainty in Archaeological Computational Modeling*. Forum, Moderated by Marieka Brouwer Burg. Society for American Archaeology, 79th Annual Meeting, Austin, Texas.
- 2012 Chair, *What Role did Changing Temperatures Play in the Success of North American Farming Societies in the Last Prehispanic Millennium?* Symposium, Society for American Archaeology, 77th Annual Meeting, Memphis, Tennessee.
- 2011 Participant, *Graduate Workshop in Computational Social Science*. Directed by John Miller and Scott Page. Santa Fe Institute, Santa Fe, New Mexico.
- 2010 Co-organizer, *The Village Ecodynamics Project II*. Symposium, Society for American Archaeology, 75th Annual Meeting, St. Louis, Missouri.

Academic Service

Peer-review

American Antiquity — 2019, 2016
Frontiers in Earth Science — 2021
Journal of Anthropological Archaeology — 2019
Journal of Archaeological Method and Theory — 2022, 2018, 2016, 2015
Journal of Archaeological Science — 2020, 2017, 2016
Journal of Archaeological Science: Reports — 2021, 2020, 2015
Journal of Ethnobiology — 2015
KIVA: Journal of Southwestern Anthropology and History — 2022, 2020

PLoS ONE — 2018
Proceedings of the National Academy of Sciences — 2019

Panel Service
 NOAA National Integrated Drought Information System — 2024
 National Science Foundation — 2023

Proposal review
Graduate Women in Science — 2020
 National Science Foundation — 2022, 2019

National Climate Assessment
 2011–2012 Board member, The National Science Foundation and TERC

Society for American Archaeology
 2023–present Chair, Committee on Awards and Scholarships
 2019–present Member, Bylaws Committee
 2019–2022 Chair, Investment Committee
 2020, 2021 Member, Program Committee
 2010 Organizer, *Exploratory Meeting for the Creation of an LGBT Archaeologists Interest Group*,
 75th Annual Meeting, St. Louis, Missouri

University of Montana
 2023 Member, Scholarship Committee

Department of Society and Conservation, University of Montana
 2025 Member, Search committee, Teaching Professor in Geographic Information Systems
 (Department of Forest Management)
 2023–2025 Committee chair, Masters — Paige Johnson (Degree granted Summer 2025)
 2023–2024 Department Representative, University Faculty Association
 2023–2024 Committee member, Masters — Sophie Morris
 2019–2022 Committee member, Doctoral — Ada Smith (Degree granted December 2022)

Montana Climate Office, University of Montana
 2018 Mentor, Baccalaureate — Carly Kuske, Montana Drought & Climate project intern
 2018 Mentor, Baccalaureate — Colin Brust, Montana Space Grant Consortium intern

Crow Canyon Archaeological Center, Cortez, Colorado
 2019–2020 Mentor, Post-doctoral — Michelle Turner

Department of Anthropology, University of Colorado, Boulder
 2019–2022 Mentor, Doctoral — Kaitlyn Davis

Department of Anthropology, Washington State University
 2020–2024 Committee member, Doctoral — Darcy Bird
 2018–2023 Mentor, Doctoral — Andrew Gillreath-Brown
 2013 Co-organizer, Interdisciplinary Research in Anthropology and the Sciences Conference
 2011 Chair, Banquet Committee, Anthropology Graduate Organization
 2009 Chair, Lipe Scholar Committee, Anthropology Graduate Organization
 2008–2009 President, Anthropology Graduate Organization

Whole Tale: Computing environments for reproducibility
 2017 Mentor, Whole Tale Summer Internship, Chicago, Illinois — Pratik Shrivastava

National Science Foundation IGERT.org Trainee Advisory Board
 2011–2012 Board member, The National Science Foundation and TERC

Outreach, Press, & Broader Impacts

anti/normal: 10,000 years of diversity, adaptation, and ecological resilience

Invited lecture

- **2025-08-14** Indigo Partners annual meeting.

Geographic Imaginings in Conservation Social Science

Guest lecture (classroom)

- **2024-10-23** Invited lecture, Conservation Social Science (NRSM 500, Prof. Hilary Faxon).

Enhancing Sustainable Disaster Relief in FSA Programs

Public data

- **2025 FOIA Archive:** *Annual County Eligibility Data from the Livestock Forage Disaster Program, 2008–2024*. FOIA request 2025-FSA-08422-F fulfilled August 7, 2025. This replaces previously filed and fulfilled FOIA response covering 2012–2022 (2023-FSA-00937-F) and 2008–2024 (2025-FSA-04690-F).
HTML: <https://sustainable-fsa.github.io/fsa-lfp-eligibility/>
- **2025 FOIA Archive:** *Farm Service Agency Normal Grazing Period Archive, 2008–2024*. FOIA request 2025-FSA-04691-F fulfilled April 15, 2025.
HTML: <https://sustainable-fsa.github.io/fsa-normal-grazing-period/>
- **2025 FOIA Archive:** *Livestock Forage Program Counties – The county boundary dataset used to determine LFP eligibility from the USDA Farm Service Agency, 2008–2024*. FOIA request 2025-FSA-08431-F fulfilled September 8, 2025.
HTML: <https://sustainable-fsa.github.io/fsa-lfp-counties/>
- **2025 Archive:** *Weekly US Drought Monitor (USDM) data archived as GeoParquet files with detailed metadata, January 4, 2000 – Today*.
HTML: <https://sustainable-fsa.github.io/usdm/>
- **2025 Derived Data:** *Weekly US Drought Monitor (USDM) data aggregated to US Census county boundaries*.
HTML: <https://sustainable-fsa.github.io/usdm-counties/>

Native Drought Resilience: CSKT Actions for Climate-Drought Adaptation

Public lecture

- **2025-04-10** *Native Drought: CSKT Actions for Climate-Drought Adaptation*, 2025 Crown Managers Partnership Forum, *Interweaving Indigenous Knowledge and Western Science with Collaborative Conservation in the Crown of the Continent*, Pablo, Montana.
- **2022-12-13** *Interweaving Traditional Knowledge and mainstream science to address long-term drought resilience priorities of the Confederated Salish and Kootenai Tribes*, The Wilderness Society.

Press coverage

- **2023-06-12** New Mesonet Station Records & Delivers Climate Data at the Confederated Salish and Kootenai Tribes Bison Range. *Drought.gov*. 12 June 2023. [html]
- **2023-05-30** New Mesonet station records and shares weather data at CSKT Bison Range. *KPAX (CBS)*. [html]
- **2023-05-25** New Mesonet station will bring up to date weather information to the CSKT Bison Range. *Charkoosta News*. [html]
- **2023-05-25** New Meteorological Station Delivers Real-time Climate Data to CSKT Bison Range. *Flathead Beacon*. [html]
- **2023-05-22** Weather tower to go in on CSKT Bison Range. *NBC Montana*. [html]
- **2023-05-17** New Montana Mesonet Station Installed on the CSKT Bison Range. *Native Climate*. [html]

Native Resilience: Strengthening the role of Climate Hubs in Indian Country

Public data

- **2022** *Native Climate Tribal College CMIP6 Agricultural Climate Projections*. Downscaled projections for 35 Tribal colleges and universities across the contiguous US. A product of the Native Climate Project. [html]

Guest lecture (classroom)

- **2025-04-01** *Telling Stories of Agri-Futures: The Native Resilience Tribal Ag Projects*, invited lecture, Global Environmental Politics (PSCI 332, Prof. Karen Adams).
- **2023-07-25** *Strengthening the Role of the Climate Hubs in Indian Country*, invited lecture, Environmental Biology (BIO 105, Prof. Mark Clytus).
- **2023-07-13** *Wildfire Smoke Resilience Workshop*, TRIO Upward Bound, University of Montana.
- **2023-04-26** *Strengthening the Role of the Climate Hubs in Indian Country*, invited lecture, Environmental Biology (BIO 105, Prof. Mark Clytus).
- **2023-03-27** *Climate Data in R*, invited lecture, GEOG 701, University of Nevada, Reno (Prof. Christine Albano).

Public lecture

- **2025-09-10** *Living in a Changing World: Drought Briefing for the United States*, Southwest Drought Learning Network drought briefing webinar.
- **2025-06-17** *Living in a Changing World: Climate Outlooks for the United States*, Native Climate Working Group webinar.
- **2025-06-10** *Tribal Drought Update & Resources: Columbia and Upper Missouri River Basins*, NOAA NIDIS webinar.
- **2025-03-18** *Living in a Changing World: Climate Outlooks for the United States*, Native Climate Working Group webinar.
- **2025-01-07** *Telling the stories of climate futures — The Native Climate Tribal Ag Projections*, USDA NIFA Tribal webinar.
- **2024-09-17** *Living in a Changing World: Climate Outlooks for the United States*, Native Climate Working Group webinar.
- **2024-05-29** *Nevada Drought Update*, Nevada Tribal Ag and Food Summit.
- **2024-05-08** *Agriculture Risk in a Changing World: Climate Outlooks for the United States*, Climate Action for Agriculturalists short course, Farmer Campus.
- **2024-04-19** *Living in a Changing World: Climate at the Crown of the Continent*, CSKT Earth Action Day, Pablo, MT.
- **2024-04-09** *Agriculture Risk in a Changing World: Climate Outlooks for the United States*, 2024 ERME National Conference, Salt Lake City, UT.
- **2024-02-20** *Western Drought Update*, Native Climate Working Group webinar.
- **2023-11-28** *Living in a Changing World: Climate at the Crown of the Continent*, Regional Climate Symposium, Polson, MT.
- **2023-11-28** *Living in a Changing World: Climate at the Crown of the Continent*, Regional Climate Symposium, Polson, MT.
- **2023-09-26** *Drought in a Changing World: The Past, Present, and Future of Drought Adaptation and Resilience in the Columbia and Missouri River Basins*, Workshop for Building Drought Resilience in a Changing Climate with Upper Columbia and Missouri Basin Tribes.
- **2023-09-19** *Western Drought Update*, Native Climate Working Group webinar.
- **2023-08-23** *Western Drought Update*, Native Climate Working Group webinar.
- **2023-06-20** *Western Drought Update*, Native Climate Working Group webinar.
- **2023-03-17** *Western Drought Update*, Native Climate Working Group webinar.
- **2023-03-14** *Observed Trends and Climate Outlook: A review of Montana's past, present, and future*, AgroClimate Workshop for Agricultural & Forestry Professionals and Partners, Billings, MT.
- **2023-02-06** *Observed Trends and Climate Outlook: A review of Montana's past, present, and future*, Mountain Rain or Snow Workshop, Missoula, MT.
- **2022-09-20** *Western Drought Update*, Native Climate Working Group webinar.
- **2022-06-28** *Nevada Drought Update*, Nevada Tribal Ag and Food Summit.
- **2022-06-23** *AZ Drought Update and FRTEP Drought Monitoring Program*, AZ Federally Recognized Tribes Extension Program monthly call.
- **2022-05-13** *Western Drought Update*, Native Climate Working Group webinar.

Mentorship

- **2023** *All Climate is Local*. Mentored 20 students through their ACL projects; recorded and

edited their final presentations. [html]

Drought Assessment has been Outpaced by Climate Change: Empirical Arguments for a Paradigm Shift

Public lecture

- **2023-04-04** *The Futures and Fortunes of Drought Relief under Climate Change*, University of Montana WA Franke College of Forestry and Conservation, Conservation Social Science Seminar.
- **2022-10-06** NOAA National Integrated Drought Information System Executive Committee meeting, Washington, DC.
- **2022-09-13** Webinar and discussion, Drought Monitoring in a Changing Climate.

Sustainable Agricultural Systems in the Past, Present, and Future

2023-02-03 *Reconstructing Climates of the Past*. Radio/Podcast interview. Science Moab. [html]

Guest lecture (classroom)

- **2022-11-07** *Visions of Desert Futures in Arid Pasts*, University of Montana WA Franke College of Forestry and Conservation, Climate & Society.
- **2022-11-03** *Visions of Desert Futures in Arid Pasts*, Harvard University Graduate School of Design, Desert Futures.

Public lecture

- **2020-01-24** Crow Canyon Webinar Series.
- **2019-08-12** Four Corners Lecture Series, Cortez, Colorado. Invited by the Hisatsinom Chapter of the Colorado Archaeological Society.
- **2016-05-21** In Smithsonian Associates Weekend Program: *Vanished! Behind the Mystery of Mesa Verde*, S. Dillon Ripley Center, Washington, DC.

Experimental Farming: The Pueblo Farming Project and Beyond

Public lecture

- **2017-04-22** In Crow Canyon Research Seminar: *Connecting Mesa Verde and the Rio Grande through the Study of Pueblo Foodways*, New Mexico History Museum, Santa Fe, NM.

Origins: The Journey of Humankind

2017-04-17 Featured expert, The National Geographic Channel. Featured in Season 1, Episode 6: *Building the Future*. Aired April 17, 2017. [html].

Exploration and exploitation in the macrohistory of the pre-Hispanic Pueblo Southwest

Popular article

- **2016-04-06** Do prehistoric Pueblo populist revolutions presage American politics today? Kyle Bocinsky and Tim Kohler, *The Conversation*. [html]

Press coverage

- **2016-04-08** Climate and the Collapse of Ancient Civilizations. Interview on NPR's *Science Friday*. [html]
- **2016-03-31** From drought, seeds of discontent. Staci Matlock, *Santa Fe New Mexican*. [html]
- **2016-04-01** Social collapse in ancient Pueblo civilization linked to climate change. Hannah Osborne, *International Business Times*. [html]
- **2016-04-10** Researchers: Four Corners pueblo culture sensitive to climate. Emery Cowan, *Arizona Daily Sun*, Flagstaff, Arizona. [html]

Can Pueblo Corn Save Ethiopian Farms? Employing 1,400 Years of Agricultural Knowledge in Service of the Future

Public lecture

- **2016-07-25** Southwest Seminar, Santa Fe, New Mexico.
- **2016-01-18** Arizona Archaeological and Historical Society, Tucson, Arizona.
- **2015-08-04** Four Corners Lecture Series, Cortez, Colorado. Invited by the Hisatsinom Chapter of the Colorado Archaeological Society.

FedData: Functions to Automate Downloading Geospatial Data Available from Several Federated Data Sources

Blog coverage

- **2016-03-15** Using the new R package, FedData, to access federal open datasets (including interactive graphics). ZevRoss.[html]

**Travel,
Outreach, &
Training Grants**

The Crow Canyon Ochre Project: Building Online Communities for Heritage Research and Education
2020 Outreach grant, Colorado Humanities CH-CARES program, \$5,250 awarded to CCAC

Toward Cultural Models of the Development of Ancestral Maize Landraces
2014 Travel grant, Graduate & Professional Student Association, Washington State University, \$375

Understanding and modeling turkey use in the American Southwest: A preliminary model for RePAST

2009 Travel grant, Graduate & Professional Student Association, Washington State University, \$390

Archaeological Field School with Brooklyn College, City University of New York

2007 Travel grant, Undergraduate Research Opportunity Program, University of Notre Dame, \$1,000

2007 Travel grant, Glynn Family Honors Program, University of Notre Dame, \$1,500

2007 Learning Beyond the Classroom Grant, University of Notre Dame, \$711

The similarity of treatment needs between the chemically addicted and HIV+ populations: Emphasizing relationships

2007 Travel grant, Undergraduate Research Opportunity Program, University of Notre Dame, \$1,000

2006 Americorps Service Grant, \$1,000

**Additional
Training**

Beyond Archaeology—Realities of Cultural Resource Management

2013 Participant in seminar presented by the Confederated Tribes of the Colville Reservation History/Archaeology Program. October 4–5, 2013, Pullman, Washington.

**Courses
Taught**

University of Montana, Missoula, Montana

Spring 2024 Instructor, *FORS 350 / GPHY 488: (Forestry) Applications of GIS*, 3 credits, 51 students

An advanced GIS course in thematic mapping, offered as part of the Certificate in GIS

Spring 2023 Co-facilitator (with Jennifer Thomsen), *NRSM 594: Conservation Social Science Seminar*, 1 credit, graduate

Presentations by student, faculty, and associates on issues and topics in their field

Washington State University, Pullman, Washington

Fall 2015 Co-instructor (with Edward Hagen), *ANTH 600: Exploratory Data Analysis in R*, 3 credits, graduate

Using graphical and numeric computational methods to explore different types of data, including survey, experimental, textual, spatial, and network data

Fall 2014 Instructor, *ANTH 101: General Anthropology*, 3 credits, 90 students

An introduction to the four fields of anthropology

Spring 2013 Instructor, *ANTH 490: Themes in Anthropology*, 3 credits, 15 students

The capstone seminar in the anthropology major at WSU; a review of important themes in anthropological thought and several controversial issues in anthropology

Spring 2012 Instructor, *ANTH 331: The Americas Before Columbus*, 3 credits, 40 students
A review course covering North and South American prehistory, and emphasizing model-based approaches in archaeology

**Professional
Affiliations**

Register of Professional Archaeologists, Registrant ID 46729296
Society for American Archaeology
American Association for the Advancement of Science